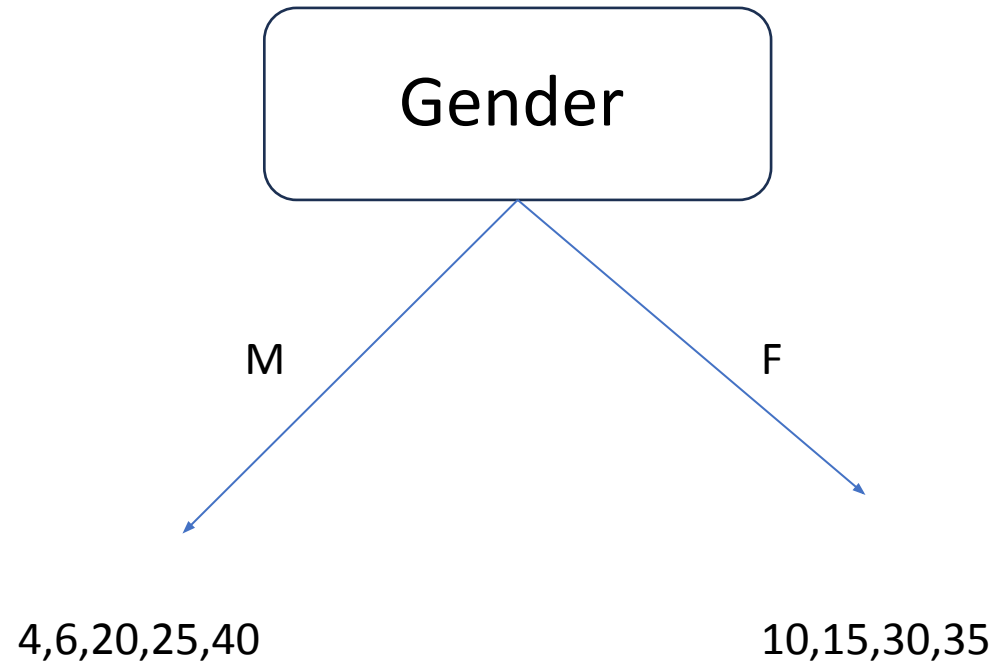


Step by Step Procedure for Regression Tree Construction

Dosage	Gender	Drug Effectiveness
4	M	0
6	M	2
10	F	3
15	F	100
20	M	100
25	M	100
30	F	20
35	F	10
40	M	0



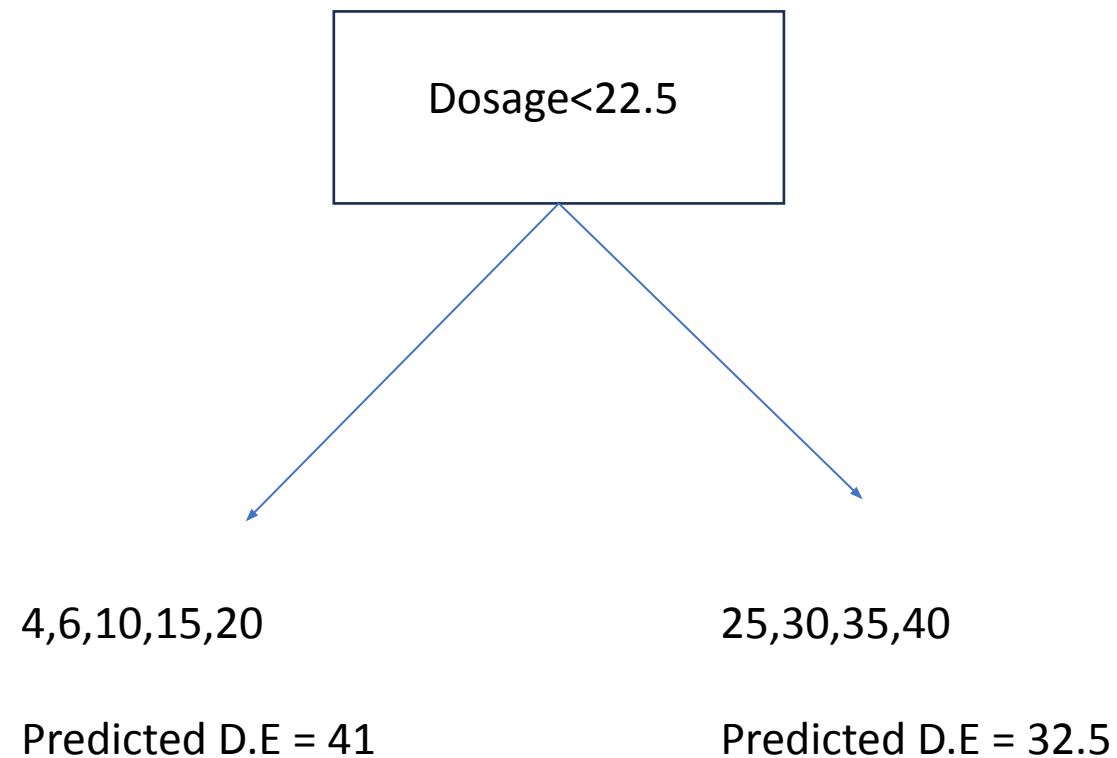
$$\text{Predicted D.E} = (0+2+100+100+0)/5 \\ = 40.4$$

$$\text{Predicted D.E} = (3+100+20+10)/4 \\ = 33.25$$

$$\text{SSR (Gender)} = (0-40.4)^2 + (2-40.4)^2 + (100-40.4)^2 + (100-40.4)^2 + (0-40.4)^2 \\ + (3-33.25)^2 + (100-33.25)^2 + (20-33.25)^2 + (10-33.25)^2 \\ = 17929.95$$

Dosage	Gender	Drug Effectiveness
4	M	0
6	M	2
10	F	3
15	F	100
20	M	100
25	M	100
30	F	20
35	F	10
40	M	0

Threshold
5
8
12.5
17.5
22.5
27.5
32.5
37.5



$$\begin{aligned}
 \text{SSR (Dosage} < 22.5) &= (0-41)^2 + (2-41)^2 + (3-41)^2 + (100-41)^2 + (100-41)^2 \\
 &\quad + (100-32.5)^2 + (20-32.5)^2 + (10-32.5)^2 + (0-32.5)^2 \\
 &= 17883
 \end{aligned}$$

Dosage	Gender	Drug Effectiveness
4	M	0
6	M	2
10	F	3
15	F	100
20	M	100
25	M	100
30	F	20
35	F	10
40	M	0

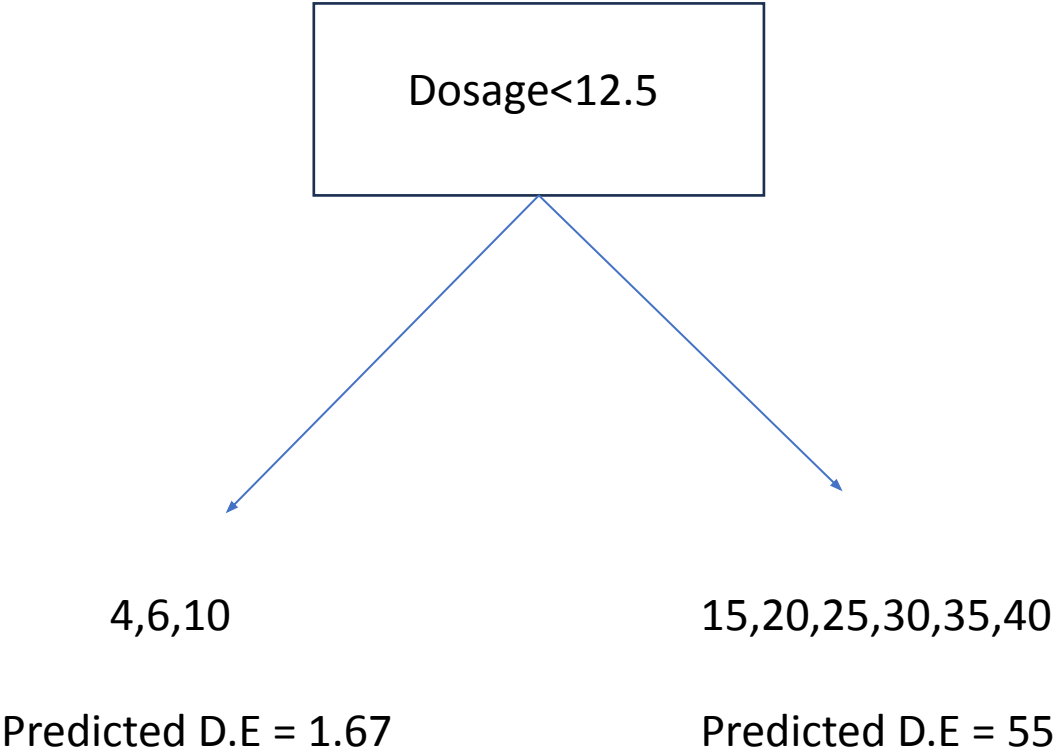
Threshold
5
8
12.5
17.5
22.5
27.5
32.5
37.5

SSR
16485
14670
12355
17177
17883
14709
15374
16485

Dosage	Gender	Drug Effectiveness
4	M	0
6	M	2
10	F	3
15	F	100
20	M	100
25	M	100
30	F	20
35	F	10
40	M	0

Threshold
5
8
12.5
17.5
22.5
27.5
32.5
37.5

SSR
16485
14670
12355
17177
17883
14709
15374
16485



Dosage	Gender	Drug Effectiveness
4	M	0
6	M	2
10	F	3
15	F	100
20	M	100
25	M	100
30	F	20
35	F	10
40	M	0

Threshold

17.5

22.5

27.5

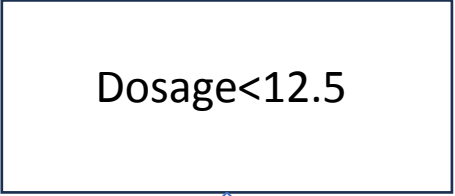
32.5

37.5

SSR

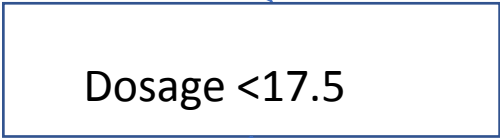
9920

6275



4,6,10

Predicted D.E = 1.67



15

Predicted D.E = 100

20,25,30,35,40

Predicted D.E = 46

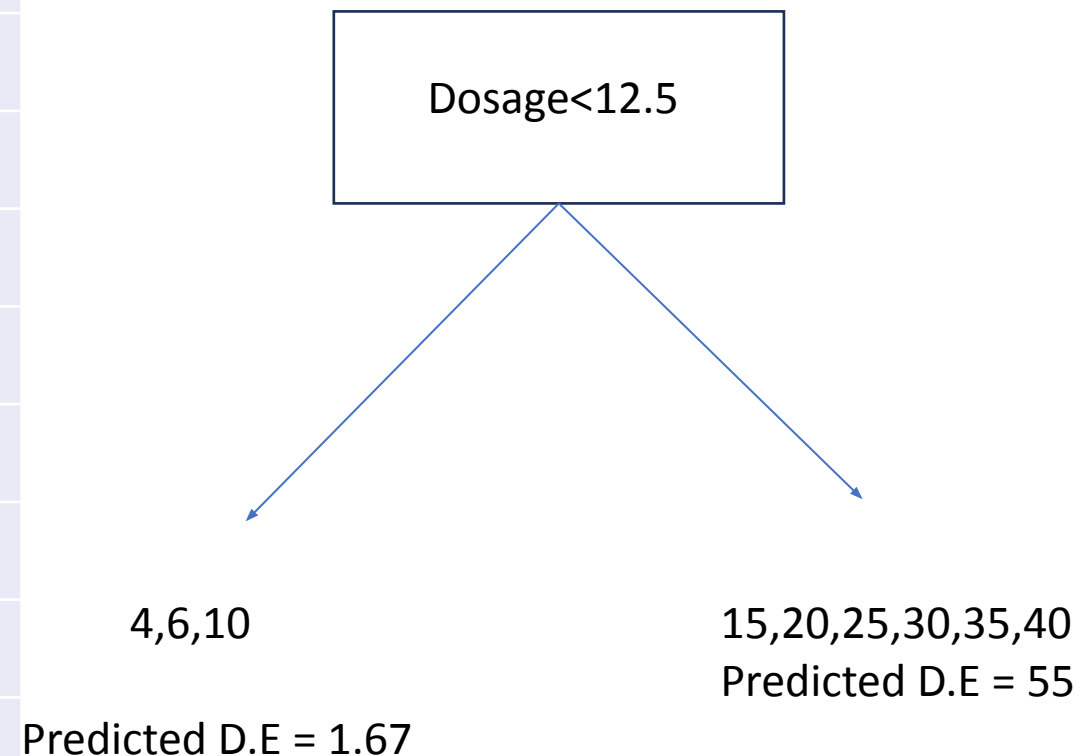
$$\text{SSR}(\text{Dosage} < 17.5) = (100-100)^2 + (100-46)^2 + (100-46)^2 + (20-46)^2 + (10-46)^2 + (0-45)^2 = 9920$$

$$\text{SSR}(\text{Dosage} < 22.5) = (100-100)^2 + (100-100)^2 + (100-32.5)^2 + (20-32.5)^2 + (10-32.5)^2 + (0-32.5)^2 = 6275$$

Dosage	Gender	Drug Effectiveness
4	M	0
6	M	2
10	F	3
15	F	100
20	M	100
25	M	100
30	F	20
35	F	10
40	M	0

Threshold
17.5
22.5
27.5
32.5
37.5

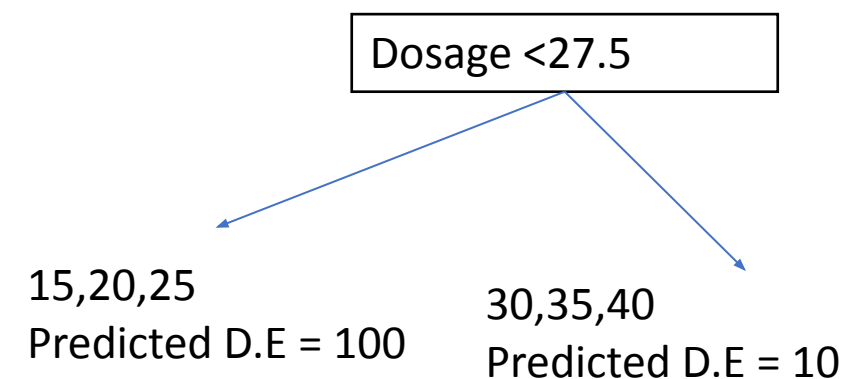
SSR
9920
6275
200
7425
13076



$$\begin{aligned} \text{SSR}(\text{Dosage} < 27.5) &= \\ (100-100)^2 + (100-100)^2 + (100-100)^2 + (20-10)^2 + (10-10)^2 + (0-10)^2 \\ &= 200 \end{aligned}$$

$$\begin{aligned} \text{SSR}(32.5) &= (100-105)^2 + (100-105)^2 + (100-105)^2 + (20-105)^2 \\ &+ (10-5)^2 + (0-10)^2 = 7425 \end{aligned}$$

$$\begin{aligned} \text{SSR}(37.5) &= (100-66)^2 + (100-66)^2 + (100-66)^2 + (20-66)^2 + \\ &+ (10-66)^2 + (0-66)^2 = 13076 \end{aligned}$$



Dosage	Gender	Drug Effectiveness
4	M	0
6	M	2
10	F	3
15	F	100
20	M	100
25	M	100
30	F	20
35	F	10
40	M	0

Threshold

17.5

22.5

27.5

32.5

37.5

SSR

9920

6275

200

7425

13076

Dosage < 12.5

4,6,10

Predicted D.E = 1.67

15,20,25,30,35,40

Predicted D.E = 55

Gender

M

20,25,40

Predicted D.E = 66.7

F

15,30,35

Predicted D.E = 43.3

SSR(Gender) =

$$(100-66.7)^2 + (100-66.7)^2 + (0-66.7)^2 + (100-43.3)^2 + (20-43.3)^2 + (10-43.3)^2 = 6666.67 + 3214.89 + 542.89 + 1108.89$$

$$= 10990.45$$

Hence among Gender and Dosage < 27.5, Dosage < 27.5 is the winner

Dosage	Gender	Drug Effectiveness
4	M	0
6	M	2
10	F	3
15	F	100
20	M	100
25	M	100
30	F	20
35	F	10
40	M	0

